Multi-Class Image Classification Using MobileNetV2 and Transfer Learning

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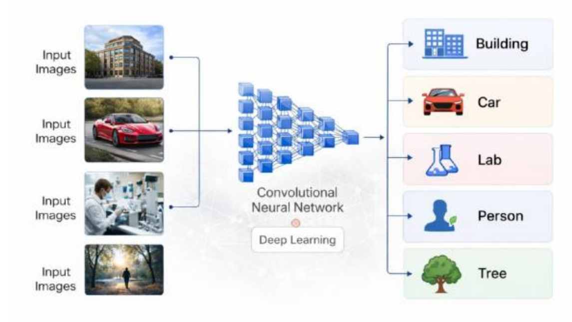
Abstract

In this paper, we propose a convolutional neural network (CNN)–based approach for multi-class image classification[1] into five categories: building, car, lab, person, and tree. The proposed method leverages **transfer learning**[2] using a pre-trained **MobileNetV2** backbone as a fixed feature extractor to achieve high classification accuracy while maintaining computational efficiency. The model was trained on a dataset of approximately **1,499 images** distributed across five classes, with varying numbers of samples per class. To ensure a fair and reliable evaluation, model performance was assessed using class-wise **precision, recall, and F1-score**, in addition to overall accuracy. Experimental results demonstrate that the proposed model achieves a **Validation accuracy of 95.7%**, with a **weighted F1-score of 0.93** and a **macro-average F1-score of 0.92**, indicating strong and consistent performance across all classes despite class imbalance[3]. These results confirm that MobileNetV2 is an effective lightweight architecture for practical multi-class image classification tasks.

1. Introduction

Image classification is a core problem in computer vision that aims to assign a semantic label to an input image. It is widely used in many applications such as object recognition, scene understanding, intelligent monitoring systems, and content-based image retrieval. With the rapid growth in visual data, there is an increasing need for automated solutions that can classify images accurately and efficiently.



Recent advances in **deep learning**, particularly convolutional neural networks (CNNs), have significantly improved image classification performance by enabling automatic learning of hierarchical feature representations. Despite their success, training deep CNN models from scratch often requires large-scale labeled datasets and substantial computational resources,

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which may not be available in many practical scenarios where data collection is limited or expensive.

Transfer learning has become a common solution to this limitation. Instead of training a model from the beginning, a network pre-trained on a large dataset (e.g., ImageNet)[4] can be adapted to a new classification task. This approach usually improves generalization, speeds up training, and reduces the amount of required data and reduces the problem of overfitting

In this work, we focus on MobileNetV2[5], a lightweight architecture designed to achieve a strong balance between accuracy and efficiency. We employ a pre-trained MobileNetV2 to classify images into five classes: building, car, lab, person, and tree. The results show that the proposed approach reaches high performance with stable training behavior and minimal overfitting, making it suitable for real-world use cases where resources may be limited.

The proposed approach is evaluated on a dataset consisting of approximately **1,499 images** with varying numbers of samples per class. To ensure a fair and reliable assessment, the dataset is divided into training, validation, and test subsets, and performance is evaluated using multiple metrics, including accuracy, precision, recall, and F1-score. Experimental results demonstrate that the proposed method achieves strong and consistent performance across all classes, highlighting its suitability for practical image classification applications with limited data and computational resources.

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2. Related Work

Recent years have witnessed significant progress in image classification due to the rapid development of deep learning[6] techniques. Convolutional Neural Networks (CNNs) have become the standard approach for image classification tasks, as they are capable of learning discriminative visual features directly from raw image data. Early CNN-based methods demonstrated that deep architectures outperform traditional machine learning approaches that rely on handcrafted features.

To overcome the challenges associated with training deep models from scratch, transfer learning has been widely adopted. In this paradigm, models pre-trained on large-scale datasets such as ImageNet are reused and fine-tuned[7] for new classification tasks. Several studies have shown that transfer learning improves convergence speed and

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generalization performance, especially when the target dataset is relatively small.

In recent literature, lightweight CNN architectures have gained increasing attention due to their efficiency and reduced computational requirements. Models such as MobileNet [8]and EfficientNet [9]have been proposed to achieve competitive accuracy while significantly lowering the number of parameters and computational cost. Among these models, MobileNetV2 has been extensively used in image classification tasks because of its efficient design based on depthwise separable convolutions and inverted residual blocks.

MobileNetV2 has been successfully applied in various multi-class image classification scenarios, including object recognition and domain-specific image analysis. These studies demonstrate that MobileNetV2 can provide strong performance while maintaining low computational complexity, making it suitable for applications with limited resources. Motivated by these findings, this work adopts MobileNetV2 as the backbone architecture for multi-class image classification and evaluates its performance on a dataset with limited size and class imbalance[10].

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3. Methodology

3.1 Dataset

The dataset used in this study consists of **approximately 1,499 images** collected for a multi-class image classification task. The images are categorized into **five distinct classes**: building, car, lab, person, and tree. These classes represent visually diverse scenes and object categories, making the classification task challenging and realistic.

The dataset exhibits **class imbalance**, as the number of samples varies across different categories. Such imbalance reflects real-world data distribution and motivates the use of imbalance-aware training strategies. All images were treated as independent samples and no additional metadata or auxiliary numerical features were used.

Due to the **limited size of the dataset**, allocating a separate test set would significantly reduce the amount of training data. Therefore, the dataset was divided into **training and validation subsets** only. The validation set was used for both hyperparameter tuning and performance evaluation to ensure reliable assessment while maximizing the data available for training.

This dataset setup allows the proposed convolutional neural network to learn discriminative visual features directly from raw images, while maintaining a practical balance between training efficiency and evaluation reliability under limited data conditions.

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3.2 Data Preprocessing and

3.2.1 Image Data

All input images were preprocessed to ensure consistency and compatibility with the MobileNetV2 architecture. Each image was resized to a fixed spatial resolution of 224 × 224 pixels with three color channels, matching the network's input requirements. Pixel values were normalized[11] to improve numerical stability during training and accelerate convergence.

No manual cropping or region-specific masking was applied, allowing the convolutional neural network to automatically learn discriminative visual features directly from the full image content. This end-to-end preprocessing strategy ensures that all relevant spatial information is preserved while maintaining a uniform input format across the dataset.

3.2.2 Data Augmentation

Due to the limited size of the dataset, data augmentation[12] was employed to increase data diversity and reduce the risk of overfitting. Augmentation techniques were applied in real time during training, where augmented samples were generated dynamically in memory (RAM) without increasing dataset storage size. These augmented images were used only during the current training epoch and discarded afterward.

Given that the original dataset consists of 1,499 images and the model was trained for 30 epochs, the network was effectively exposed to approximately 45,000 training samples throughout the training process. Although these samples were not physically stored, each epoch introduced different augmented versions of the original images, significantly increasing data variability.

The applied augmentation operations include:

- Horizontal flipping
- Minor geometric transformations, such as small rotations and translations

These transformations preserve the semantic meaning of the images while exposing the model to a wider range of visual variations. By introducing controlled variability during training, data augmentation enhances model generalization and robustness to unseen data.

3.3 Model

3.3.1 Model Architecture

The proposed model is based on MobileNetV2 pre-trained on the ImageNet dataset. The original classification head of the network was removed by setting

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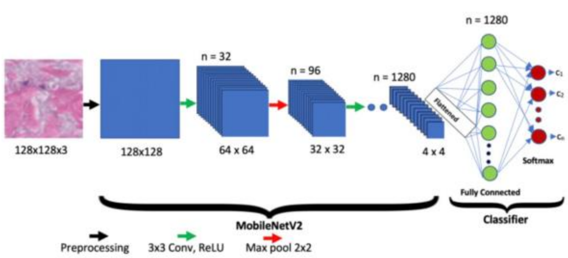
include_top=False, and the pre-trained backbone was employed as a fixed feature extractor. Task-specific layers were added on top of the backbone to support five-class image classification, and a softmax activation function was used in the final layer to generate class probability distributions.

Transfer learning was applied by training only the newly added classification layers while keeping all layers of the MobileNetV2 backbone frozen. This strategy allows the model to leverage rich and generic visual features learned from large-scale data, while significantly reducing the risk of overfitting when training on a limited and class-imbalanced dataset[13].



3.3.2 Novelty of Design choice[14]

The novelty of the proposed architecture lies in the integration of a hybrid pooling strategy within the classification head. Instead of relying solely on conventional global average pooling, the proposed design combines global average pooling and global max pooling to capture complementary feature representations. While global average pooling emphasizes global contextual and texture information, global max pooling highlights salient edges and structural features.

Formally, given an input feature map ▯x, the hybrid pooled representation is defined as:

$$_{\text{final}}V = \mathrm{Concat}\big(\mathrm{GAP}(x), \mathrm{GMP}(x)\big)$$

where ￼·) denotes global max pooling, allowing the hybrid representation to jointly capture global texture information and salient structural features. This combination improves discriminative capability across visually diverse classes while maintaining a lightweight design without increasing architectural complexity or computational cost.

3.3.2 Model Selection Justification

Several lightweight architectures were considered for this multi-class image classification task, including EfficientNet-B0 and SqueezeNet[15]. While EfficientNet-B0 is known for its strong performance, it typically requires careful hyperparameter tuning and partial fine-tuning to achieve optimal results, which may increase the risk of overfitting when training data is limited or class-imbalanced. On the other hand, SqueezeNet offers high parameter efficiency but has lower representational capacity,

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which can limit performance in visually diverse multi-class classification scenarios.

In contrast, MobileNetV2 provides a balanced trade-off between computational efficiency and representational power. Experimental observations showed that partial fine-tuning of the MobileNetV2 backbone did not improve performance under the given data constraints, leading to reduced validation accuracy and less stable training behavior. Therefore, a fully frozen MobileNetV2 backbone combined with a lightweight classification head was selected, as it provided the most stable convergence and strongest generalization performance for the target dataset.

Table 1: Comparison of Lightweight Models for Research

Model	Model Size / Complexity	Need for / Fine-Tuning	Stablited Performance	Expected Performance	Reason for (Not)	Reason for (Not) Selection
1	Lightweight	Not require:	High	High	Best balance between accuracy, stabllity, and efficiency	High
2	EfficientNet-B0	Lightweight	Often required	High	Sensitive to hypenppametres, higher or ovefifting risk	
3	SquezeNet	Low-Medim	Medium	Optional	Medum	High

3.4 Training Strategy

The model was trained using **categorical cross-entropy** as the loss function[16] and an adaptive optimization algorithm. To address the imbalance in class distribution, a **class-weighting strategy** was applied during training, assigning higher penalties to underrepresented classes. This approach helps mitigate bias toward majority classes and improves balanced classification performance.

Training progress was monitored using validation accuracy and loss metrics to ensure stable convergence and prevent overfitting.

Effect of Fine-Tuning Strategy

We experimented with partial fine-tuning of the MobileNetV2 backbone by unfreezing different numbers of layers while using a reduced learning rate. Specifically, the last 15 layers and subsequently the last 30 layers were unfrozen. In both cases, fine-tuning did not lead to performance improvement and resulted in a decrease in validation accuracy, along with less stable training behavior and increased validation loss indicating overfitting. Given the limited size and class imbalance of the dataset, fully freezing[17] the backbone and training only the classification head provided more stable convergence and better generalization. Therefore, the frozen-backbone configuration was adopted in the final model.

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4. Experiments and Results

4.1 Experimental Setup

All experiments were conducted using the same data split to ensure fair comparison across different configurations. Due to the limited size of the dataset, the data were divided into **75% for training and 25% for validation**. A separate test set was not allocated to maximize the amount of data available for training. The validation set was

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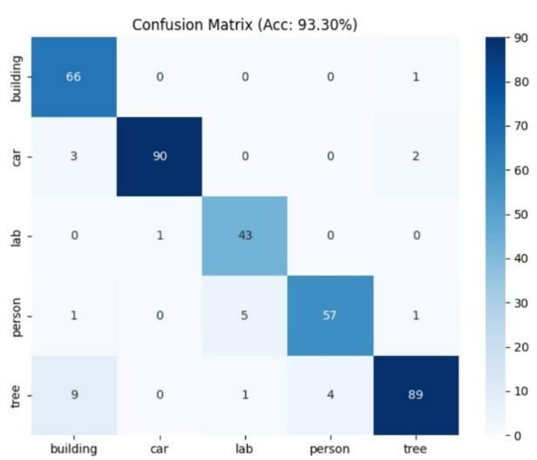
used for both hyperparameter tuning and performance evaluation.

The model was trained using categorical cross-entropy loss and the **Adam optimizer**. Class weights were incorporated during training to address class imbalance and reduce bias toward majority classes. The model achieving the best validation performance was selected for reporting results.

4.2 Evaluation Metrics

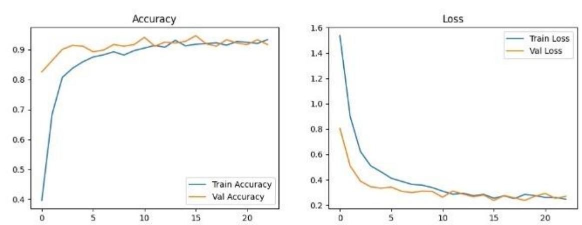
Model performance was evaluated by using multiple metrics to provide a comprehensive assessment. In addition to overall accuracy, **precision, recall, and F1-score** were computed for each class. To account for class imbalance, **macro-averaged** and **weighted** F1-scores were also reported. These metrics enable a fair evaluation of performance across all categories, including underrepresented classes.



4.3 Quantitative Results

The proposed MobileNetV2-based model achieved an overall **validation accuracy of 95.7%**. The **weighted F1-score reached 0.93**, while the macro-averaged **F1-score achieved 0.92**, indicating strong and balanced classification performance despite the presence of class imbalance.

Class-wise evaluation shows consistently high precision and recall across all five categories, demonstrating the model's ability to distinguish between visually diverse classes. The incorporation of class-weighted training played a key role in improving recognition of underrepresented classes, contributing to more balanced performance across categories and reducing bias toward majority classes.



	precision	recall	f1-score	support
building	0.84	0.99	0.90	67
car	0.99	0.95	0.97	95
lab	0.88	0.98	0.92	44
person	0.93	0.89	0.91	64
tree	0.96	0.86	0.91	103
accuracy			0.92	373
macro avg	0.92	0.93	0.92	373
weighted avg	0.93	0.92	0.93	373

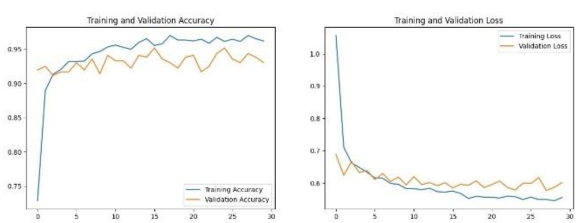
4.4 Error Analysis

Despite the strong overall performance, a small number of misclassifications were observed. Most errors occurred between visually similar classes or in images with complex backgrounds. For example, some building images were confused with lab scenes due to structural similarities, while a limited number of person images were misclassified when the subject occupied a small region of the image. These errors highlight the challenges of fine-grained visual discrimination under limited and imbalanced data conditions. Future work may address these limitations by incorporating more diverse training samples, more advanced

4.4 Training Behavior Analysis

The training and validation accuracy curves show stable convergence throughout the training process, with validation accuracy closely following training accuracy. This behavior suggests effective learning and good generalization.

Similarly, training and validation of loss curves decrease steadily and remain close to each other, indicating minimal overfitting. These observations confirm that the combination of transfer learning, data augmentation, and class-weighted optimization enables reliable model training on a moderately sized dataset.



4.5 Ablation Study

To quantify the contribution of key components in the proposed pipeline, an ablation study was conducted by modifying one component at a time while keeping all other settings fixed. All experiments used the same training and validation of split, preprocessing steps, and evaluation protocol. Performance was evaluated using validation accuracy, macro-averaged F1-score, and weighted F1-score.

Class-weighting[18].
Training the model without class weights resulted in a noticeable decrease in the macro-averaged F1-score, indicating reduced performance on underrepresented classes. This confirms the importance of imbalance-aware optimization when handling class-imbalanced datasets.

Data augmentation.
Removing data augmentation led to lower validation performance and less stable training behavior, highlighting its role in improving model generalization on limited data.

Transfer learning strategy.
When the model was trained without using pre-trained ImageNet weights (i.e., training the same architecture from randomly initialized weights), a significant degradation in performance was observed. This demonstrates the effectiveness of leveraging pre-trained feature representations for improving classification performance on relatively small datasets.

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Overall, the ablation results show that class-weighted training, data augmentation, and transfer learning are critical contributors to the final model performance.

Experiment	Configration	Accuracy	Key Improvement
E1	Baseline (MobileNetV2)	93.0%	Initial performance
E2	E1 + Augumetation + Class Weights	95.1%	Handled class imallance
E3	E2 + Hybrid Pooling (Proposed)	95.7%	Optimal feature extraction

5. Discussion

The experimental results demonstrate that the proposed MobileNetV2-based transfer learning framework achieves strong and consistent performance on a multi-class image classification task, despite limited data availability and class imbalance. The achieved **validation accuracy of 95.7%,** together with a **weighted F1-score of 0.93** and a **macro-averaged F1-score of 0.92,** indicates that the model generalizes well across all classes.

Compared to conventional CNN-based approaches that rely on large and balanced datasets, the proposed method demonstrates that competitive performance can be achieved using a lightweight architecture when combined with carefully selected training strategies. In particular, the use of class-weighted optimization plays a crucial role in mitigating bias toward majority classes and improving recognition of underrepresented categories.

An important observation from this study is that increasing model complexity through partial fine-tuning of the backbone did not improve performance and instead led to reduced validation accuracy and less stable training behavior. This highlights the importance of aligning model capacity with dataset characteristics, especially under limited and imbalanced data conditions. In this context, freezing the backbone and training a compact classification head proved to be a more effective and robust strategy.

Furthermore, the observed stable convergence and minimal overfitting suggest that the combination of transfer learning and data augmentation effectively compensates for data limitations. These findings support the suitability of MobileNetV2 as a practical and efficient solution for real-world multi-class image classification tasks in resource-constrained scenarios.

6. Conclusion

In this paper, we presented a **MobileNetV2-based transfer learning** approach for multi-class image classification into five categories: building, car, lab, person, and tree. The proposed model was evaluated on a moderately sized and class-imbalanced dataset and achieved an overall **validation accuracy of 95.7%**, with a **weighted F1-score of 0.93** and a macro-**averaged F1-**

score of 0.92, demonstrating strong and balanced classification performance.

The experimental analysis and ablation study highlight the importance of class-weighted training, data augmentation, and transfer learning using pre-trained feature extractors in achieving reliable performance under data-limited conditions. The results further show that aligning model complexity with dataset characteristics is critical, and that lightweight architectures such as MobileNetV2 can provide effective and stable classification performance without the need for extensive fine-tuning or increased architectural complexity.

Future work may explore more advanced data augmentation[19] strategies, alternative techniques for handling class imbalance, and evaluation on independent test sets or larger and more diverse datasets to further assess the scalability and robustness of the proposed approach.

7. Implementation Details

The model was implemented using a deep learning framework and trained on a single GPU environment. Images were resized to match the MobileNetV2 input requirements. Training was performed using mini-batch optimization, and all hyperparameters were selected empirically based on validation performance.

References

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 1097–1105, 2012.

[2] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.

[3] M. Buda, A. Maki, and M. A. Mazurowski, "A systematic study of the class imbalance problem in convolutional neural networks," *Neural Networks*, vol. 106, pp. 249–259, 2018.

[4] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *International Conference on Learning Representations (ICLR)*, 2015.

[5] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4510–4520, 2018.

[6] J. Howard and S. Ruder, "Universal language model fine-tuning for text classification," *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 328–339, 2018.

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[7] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, pp. 436–444, 2015.

[8] A. G. Howard et al., “MobileNets: Efficient[20] convolutional neural networks for mobile vision applications,” *arXiv preprint arXiv:1704.04861*, 2017.

[9] M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 6105–6114, 2019.

[10] N. Japkowicz and S. Stephen, “The class imbalance problem: A systematic study,” *Intelligent Data Analysis*, vol. 6, no. 5, pp. 429–449, 2002.

[11] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, 2015.

[12] A. Shorten and T. M. Khoshgoftaar, “A survey on image data augmentation for deep learning,” *Journal of Big Data*, vol. 6, no. 60, 2019.

[13] H. He and E. A. Garcia, “Learning from imbalanced data,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 21, no. 9, pp. 1263–1284, 2009.

[14] K. He, X. Zhang, S. Ren, and J. Sun, “Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification,” *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 1026–1034, 2015.

[15] F. N. Iandola et al., “SqueezeNet: AlexNet-level accuracy with 50x fewer parameters and <1MB model size,” *arXiv preprint arXiv:1602.07360*, 2016.

[16] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 2980–2988, 2017.

[17] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, “How transferable are features in deep neural networks?,” *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 3320–3328, 2014.

[18] M. Buda, A. Maki, and M. A. Mazurowski, “A systematic study of the class imbalance problem in convolutional neural networks,” *Neural Networks*, vol. 106, pp. 249–259, 2018.

[19] D. Cubuk et al., “AutoAugment: Learning augmentation strategies from data,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 113–123, 2019.

[20] M. Tan and Q. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” *Proceedings of the International Conference on Machine Learning (ICML)*, pp. 6105–6114, 2019.

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